

SOLOMON BAYE

INFORMATION SECURITY ARCHITECT

Enterprise Security Architecture | IAM & Zero Trust | Application & API Security | Cloud | Telecom & FinTech
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PROFESSIONAL PROFILE

Information Security Architect with 10 years in telecommunications, including 6 years in information security architecture at Ethio Telecom, Ethiopia's national telecom operator with 90+ million customers. Designs and assures secure architectures for digital financial services, identity and access management, Zero Trust, application and API integration, and multi-cloud environments. Translates business strategy and risk into security services, control objectives, governance models and roadmaps using SABSA, ArchiMate and TOGAF-aligned practice. CISA certified (2026), IEEE conference researcher and speaker.

10 yrs telecom | 6 yrs security architecture | 50,000+ identities protected | 15+ platforms assured | 40% assessed risk reduction*

*Contributed to people, process and technology initiatives associated with a 40% reduction in assessed enterprise risk.

PROFESSIONAL EXPERIENCE

Information Security Architect

2020 – Present

Ethio Telecom | Addis Ababa, Ethiopia | National telecommunications operator

- **telebirr Enterprise Security Architecture:** Led security architecture for a digital financial services platform serving millions of customers, using SABSA and ArchiMate to trace business drivers, security requirements, trust relationships, risks and security services through to layered controls.
- **Enterprise IAM & Zero Trust:** Architected IAM capabilities protecting 50,000+ identities, including an IBM IAM environment supporting 30,000+, covering authentication, authorization, privileged access (CyberArk PAM) and enterprise application integration; designed identity-centric access and Zero Trust controls.
- **Cloud Security:** Govern security architecture across AWS, Microsoft Azure and Huawei Cloud, assessing identity, network, workload, security posture and technology risks.
- **Solution Design & Controls:** Translate business and technical requirements into security architectures, technical specifications and architecture decisions; design and review controls including firewalls, NGFW, WAF, IDS/IPS, IAM, SIEM/SOAR, encryption and access controls across application, API, identity, data, network, infrastructure and cloud; recommend application, database, network and infrastructure design changes to meet security policies and standards.
- **Architecture Assurance:** Assess third-party solutions across 15+ major platforms (IBM IAM, QRadar SIEM/SOAR, CyberArk PAM, Radware WAF/Anti-DDoS, Huawei AAA/VDI/Cloud, WSO2, Palo Alto, Cisco, Allot DPI); review HLD, LLD, ICD and Production Acceptance Test documentation and provide security assurance before implementation and rollout, ensuring solutions meet security, regulatory and performance standards and integrate with existing applications.
- **API & Integration Security:** Review API and legacy-integration architectures, identifying application, data, network and system risks, including authentication, authorization and interface security, and delivering redesign recommendations and compensating controls.
- **Risk & Assessments:** Conduct architecture reviews, vulnerability and gap assessments and control reviews aligned to NIST CSF, ISO 27001/27005, ISO 31000 and COBIT, recommending remediation and contributing to initiatives associated with a 40% reduction in assessed enterprise risk.
- **Advisory & Leadership:** Serve as architecture assurance and audit focal point, informing technology and security investment decisions; engage and present to technical and non-technical stakeholders, vendors and executives, turning requirements and risks into practical designs; track emerging technologies and best practices; author security policies and architecture standards; deliver security training.

Network Engineer

2018 – 2020

Ethio Telecom | Addis Ababa, Ethiopia

- Engineered and hardened routing, firewall and Wi-Fi access controls supporting 99.9% availability across critical network services; deployed and managed WAF and NGFW platforms.
- Managed MSAG-to-MSAN migrations and FTTH/copper rollouts for 3,000+ customers, coordinating security planning and acceptance with Huawei vendor teams.
- Provided security design input for early enterprise cloud and hybrid network integrations, laying the foundation for the move into security architecture.

Customer Service Consultant

2016 – 2018

Ethio Telecom | Addis Ababa, Ethiopia

- First-line escalation point for complex technical incidents: performed root-cause analysis and coordinated resolution with engineering teams while maintaining 95%+ customer satisfaction.

SELECTED ARCHITECTURE PROJECTS

telebirr Enterprise Security Architecture

SABSA · ArchiMate

Ethio Telecom | Digital financial services platform serving millions of customers

- Modeled business drivers, security requirements, trust relationships, customer registration flows, security services and full-stack security layers for a mobile money environment.
- Produced an integrated trust-axis and protocol model and registration-flow security design, giving delivery and vendor teams traceable requirements from business risk to layered controls.

Enterprise IAM Architecture

IBM IAM · CyberArk PAM · Zero Trust

Ethio Telecom | 50,000+ identities protected

- Architected and assured IAM for 50,000+ identities, including an IBM IAM environment for 30,000+, covering authentication, authorization, privileged access and enterprise application integration.
- Backed by vendor-led IBM IAM training and an architecture workshop in Shanghai, China.

API Management & Integration Security

WSO2 · API Security

Ethio Telecom | Modern APIs integrated with legacy systems

- Reviewed API and legacy-integration architectures and PAT requirements across application, data, network and system security, identifying authentication, authorization and API-protection risks.
- Defined redesign recommendations and compensating controls before implementation and rollout.

Enterprise Architecture Consulting

TOGAF · Independent engagement

Ethiopian Engineering Corporation | External consulting

- Delivered an end-to-end engagement covering business and risk analysis, architecture assessment, solution recommendation and implementation sign-off, including data architecture and business drivers and principles artefacts.

SecurityBlueprint

SABSA · ISO 27001 · Independent project

Self-developed SABSA enterprise architecture and risk modeling platform

- Built a platform spanning asset discovery, gap assessment, target architecture, architecture assurance and executive reporting, structured on SABSA Layers 1–5 (Contextual to Component).
- Linked business drivers, attributes, security requirements, control objectives, implementation status and control-effectiveness scoring, with ISO 27001 Annex A traceability.
- Added automated recommendations, configurable scopes (Mobile Money, Telecom, Cloud Native, OT/ICS), Word/Excel reporting, maturity scoring, roadmaps and executive dashboards.

RECOGNITION & RESEARCH

- **Speaker, African Enterprise Security Architecture Conference (AEAC 2026):** presented practical approaches to enterprise security architecture, governance and security transformation.
- **IEEE Researcher, SEGE 2026 (Canada):** authored peer-reviewed research on IT/OT cybersecurity convergence, presented at the IEEE International Conference on Smart Energy Systems Engineering. Control Systems MSc background bridges enterprise security with critical infrastructure.
- **IBM IAM Vendor Training, Shanghai, China:** onsite vendor-led IBM IAM training and architecture-focused technical workshop.

TECHNOLOGY & CAPABILITIES

Enterprise Architecture: SABSA, ArchiMate, TOGAF-aligned architecture, target and reference architecture, security services, control objectives, roadmaps

Identity & Zero Trust: IBM IAM, IAM/IGA, CyberArk PAM, Huawei AAA, identity federation, privileged access, Zero Trust

Application & Integration: WSO2 API Management, API security, legacy integration, solution architecture, HLD/LLD/ICD

Cloud: AWS, Microsoft Azure, Huawei Cloud, hybrid cloud, security posture assurance

Cyber Defense: QRadar SIEM/SOAR, Radware WAF/Anti-DDoS, Palo Alto, Cisco, Allot DPI, NGFW, threat modeling

Governance & Risk: NIST CSF, ISO 27001/27005, ISO 31000, COBIT, gap assessment, third-party review, vendor governance

EDUCATION & CREDENTIALS

CISA – Certified Information Systems Auditor

ISACA | 2026

MSc, Electrical & Computer Engineering (Control Systems)

Addis Ababa University | 2016 – 2019

BSc, Electrical & Computer Engineering

Addis Ababa University | 2010 – 2015 | Top 5 of class